Structured Local Adaptation, Frontier Complementarity, and Metric Pitfalls in Ultra-Low-Resource Tangut Short-Text Translation

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Abstract

Tangut translation is an extreme low-resource problem: in the current repository snapshot, only 491 real Tangut-Chinese pairs are available, with 400/41/50 train-dev-test examples. The task is also structurally unusual because many targets are short, title-like Chinese strings rather than ordinary modern sentences. We therefore evaluate four method families in the same setting: local prompt-only baselines, a stronger proprietary frontier prompt-only baseline, synthetic-data supervised fine-tuning (SFT), and direct preference optimization (DPO). The new frontier baseline, DeepSeek-V3.2 with dictionary grounding and few-shot title-oriented prompting, is the strongest single non-reference system under our reference-aware judge, reaching 2.80/5 overall with 12/50 exact matches. A paper built around "stronger CoT still fails" would therefore be wrong. The positive local result instead lies inside the trainable regime: structured multitask SFT is the strongest pure supervised baseline, and stricter gap-filtered DPO on top of that base yields the strongest local model. With only 498 preference pairs whose reward gap is at least 0.4, the sigmoid variant reaches chrF++ 30.01, keeps 5/50 exact matches, preserves 22/27 title suffixes, and stays near reference length. More importantly, a best-of-two oracle between the frontier baseline and this local model reaches 13/50 exact matches, 40.16 chrF++, and 3.06 reference-aware overall, showing that local adaptation contributes complementary hypotheses rather than merely weaker copies of the frontier output. Paired bootstrap resampling confirms that the local chrF++ gain over multitask SFT is robust (+4.02, 95% CI [0.84, 7.80]), while the frontier model retains a robust advantage over the best local model on reference-aware overall (-0.58 when measured as local minus frontier, 95% CI [-1.04, -0.12]) and exact match (-0.14, 95% CI [-0.26, -0.04]). We also show that several apparently reasonable reference-free metrics are misaligned with the task: the gold human reference itself receives only moderate dictionary coverage and extremely poor perplexity under a modern-Chinese language model, despite achieving perfect chrF++ and perfect reference-aware judge scores. Finally, an overlap audit finds zero exact source, target, or pair overlap across train/dev/test splits and zero exact overlap between test references and the 50k synthetic multitask targets. These findings suggest a more precise practical lesson for ultra-low-resource Tangut short-text translation: frontier prompting is a strong proprietary upper bound, structured local supervision remains scientifically useful because it changes the error regime under severe data scarcity, and evaluation must be anchored to reference-aware metrics rather than generic fluency proxies.

1 Introduction

Machine translation for historical and ultra-low-resource languages is difficult for two separate reasons. First, parallel data is often scarce or nearly absent. Second, the outputs that matter to scholars are not always long, natural modern sentences: they may be titles, glosses, liturgical fragments, or compact bibliographic strings. Recent work on ancient-language NLP and historical-text translation has made this setting more visible, but it has also shown that data scarcity and evaluation instability are especially severe outside modern benchmark conditions [1, 2, 3]. Tangut is a canonical example. The script is historically important, but usable parallel Tangut-Chinese resources remain tiny. In the current repository snapshot, we only have 491 real Tangut-Chinese pairs.

This small-data regime also creates an obvious objection. Perhaps there is no need for task-specific adaptation at all: maybe a sufficiently strong Chinese frontier LLM, given dictionary glosses and a carefully engineered chain-of-thought-style prompt, can already do the task well enough. That objection is reasonable, so the paper should answer it directly rather than rhetorically. We therefore compare four families of methods in one frame: local dictionary-augmented prompting, a stronger proprietary frontier prompt-only baseline, synthetic pseudo-parallel SFT, and preference optimization on top of SFT [5, 4, 7]. This turns the paper into a direct test of what prompting already buys, what local training still buys, and how evaluation changes the answer.

Our main finding is now four-part. First, the stronger prompt-only objection is real: the new DeepSeek-V3.2 few-shot dictionary baseline is the strongest single non-reference system under the reference-aware judge, reaching 2.80/5 overall with 12 exact matches. Second, within the open/local trainable regime, structured synthetic supervision is still the strongest pure SFT recipe, and the strongest local systems appear only after preference optimization is moved onto that stronger base and the preference pairs are filtered much more aggressively. In particular, the strict gap-filtered DPO sigmoid variant reaches the highest local chrF++ in the repository (30.01), ties for the best local reference-aware overall score (2.22/5), keeps 5 exact matches, and stays close to reference title length. Third, the frontier comparison reveals complementarity rather than simple dominance: a best-of-two oracle between the frontier system and the best local DPO model reaches 13 exact matches, 40.16 chrF++, and 3.06 judge overall, far above either system alone. Fourth, the repository exposes two robustness issues directly: several reference-free metrics mis-rank the gold reference itself, and low-gap preference data remain noisy enough to destabilize DPO.

The repository also reveals two important failure modes. First, reference-free metrics can be badly misleading. The human reference itself is penalized by lexical coverage and perplexity, which means those metrics cannot serve as primary model-selection criteria for this task. Second, preference optimization is highly conditional. Legacy DPO on a weaker base underperforms badly, and even multitask-base DPO remains unstable when low-gap preference pairs are retained. The positive result therefore depends on both the base model and the pair-quality filter, not on DPO in the abstract.

The current draft therefore makes four contributions:

1. an empirical comparison of local prompting, frontier prompting, synthetic SFT, and DPO for Tangut short-text translation under only 491 real pairs;

2. a positive local-method result showing that structured multitask SFT is the right base and that strict gap-filtered DPO produces the strongest trainable local model, even exceeding the frontier prompt-only system on chrF++;

3. an oracle-complementarity analysis showing that the strongest local and frontier systems contribute different useful hypotheses, which motivates hybrid or scholar-in-the-loop workflows rather than a replacement narrative;

4. a reference-aware evaluation, statistical-robustness, and data-hygiene analysis showing why reference-free metrics and low-gap preference data can fail and why the present rankings are not explained by exact train/dev/test overlap.

The rest of the paper situates the task against related work, describes the experimental setting and the four comparison families, and then separates the positive local result from the frontier comparison and the metric/alignment analysis that makes the result interpretable.

2 Related Work

Ancient and historical language processing. Recent surveys argue that ancient-language NLP differs from mainstream NLP not just because data are scarce, but because scripts, editorial conventions, and annotation practices are unstable across corpora [1]. Shared tasks on ancient-language translation have likewise emphasized that even when translation is the end goal, the available supervision is often tiny and domain-specific [2]. Work on historical document modernization faces related sparsity and spelling-shift issues, but often targets orthographic or lexical modernization rather than cross-script title reconstruction [3]. Our Tangut setting is even narrower: only 491 real pairs are available, and the evaluation set is dominated by compact title-like strings.

Synthetic supervision and lexical grounding. Low-resource machine translation often relies on synthetic parallel data built from monolingual corpora, with back-translation as the canonical example [4]. A separate line of work injects lexical knowledge more directly, either by folding bilingual lexicons into the model or by constraining decoding to respect required terminology [5, 6]. Our prompting and SFT baselines sit at the intersection of these ideas: they exploit dictionary knowledge and synthetic supervision, but the central question is whether stronger gains come from input purity, approximate semantic preservation, or explicit alignment structure.

Preference optimization. Direct Preference Optimization (DPO) has become a standard offline objective for aligning generation models without a separate reward-model-plus-RL loop [7]. More recent work has pushed the same idea into machine translation, for example by using query-specific rewards for NMT or by constructing MT preferences from word alignment and MBR decoding [10, 11]. Another line of work studies robustness to noisy preference labels through modified DPO objectives or importance-style reweighting [8, 9]. Our results connect these threads in a harsher regime: the supervision is tiny, the outputs are short title-like strings, and the chosen/rejected signal is built from partially misaligned automatic metrics. In that setting, DPO can help, but only after base-model choice and pair filtering are controlled much more aggressively.

Evaluation for non-standard translation outputs. chrF++ is widely used in machine translation because character-level overlap is often more stable than token-level overlap under morphological or orthographic variation [12]. Learned evaluation and quality-estimation frameworks such as COMET and OpenKiwi push beyond surface overlap by predicting translation quality from richer signals [13, 14]. Our findings add a complementary caution: in short historical title translation, apparently reasonable reference-free metrics can still mis-rank the gold reference itself. This is why the current paper treats reference-aware evidence as primary and reference-free metrics as diagnostic.

3 Task Setting

The real Tangut data in this repository contains 491 pairs. The current split is 400 training examples, 41 development examples, and 50 test examples, produced by a single random shuffle with seed 42. Unlike ordinary machine-translation benchmarks, the targets are short and often title-like. Many examples resemble book titles, sutra titles, or compact scholastic expressions. This matters because title reconstruction rewards precise lexical choices and compact formatting, while a modern Chinese fluency model may incorrectly treat those outputs as unnatural.

Because the real dataset is tiny, the system uses synthetic data to train translation models. For each SFT branch, 50,000 synthetic Tangut-Chinese pairs are produced from monolingual classical Chinese and dictionary knowledge, then combined with the 400 real training examples upsampled by a factor of 10 to form a 54,000-example training set. All learned models use the same instruction-tuned 7B backbone.

We also explicitly audit data hygiene. There is zero exact source, target, or full-pair overlap across the train/dev/test splits, zero exact overlap between the 50 test references and the final modern-Chinese targets of the 50k synthetic multitask corpus, and zero exact overlap between the four development examples used in the frontier few-shot prompt and the test references. Character n-gram overlap is naturally non-zero inside the real dataset because titles share common Buddhist morphemes, but the synthetic multitask targets are far from copied test references: their mean maximum Jaccard overlap with test references is 0.027 for character bigrams, 0.004 for trigrams, and 0.000 for 4-grams.

We evaluate with two classes of metrics. The repository's existing metrics are lexical coverage, perplexity, chrF++, and a dictionary-conditioned LLM judge. chrF++ is a reasonable overlap anchor for this setting because character-based scoring is relatively stable under orthographic variation [12]. Perplexity is computed as $\exp(\ell)$, where ℓ is the mean token-level causal-LM loss under Qwen2.5-0.5B. In this title-heavy setting that score becomes extremely large even for the gold reference, so we treat it as diagnostic rather than dispositive. Learned MT evaluation and quality-estimation systems can capture richer signals than surface overlap [13, 14], but in this repository the available reference-free signals are only partially aligned with the task. For that reason, this draft does not treat all metrics equally: chrF++, exact match, contamination, suffix preservation, and the new reference-aware judge are primary evidence, while lexical coverage, perplexity, and the legacy reference-free judge are reported as diagnostics.

4 Methods

4.1 Inference-only baselines

The first family contains no weight updates. `baseline1` directly prompts the local base model with the Tangut string. `baseline2` augments the prompt with dictionary glosses. `baseline2_1_cot` further restructures the prompt into a three-step process intended to encourage alignment, literal drafting, and rewriting. This family is closest in spirit to lexicon-augmented or terminology-aware MT, except that the lexicon is injected through prompting rather than through decoding constraints or model-side features [5, 6].

We also add a deliberately stronger prompt-only comparison based on DeepSeek-V3.2 with few-shot dictionary prompting. This baseline is served through OpenRouter, receives the same dictionary gloss construction, uses a much stricter title-only system prompt, and is given four fixed development-set few-shot exemplars. The prompt explicitly instructs the model to prefer compact title form, avoid explanatory paraphrase, resist blindly copying generic sutra names from dictionary entries, and emit only the final title string. Decoding uses temperature 0.0 with a 512-token cap. The goal is not to build a straw baseline but to answer a reviewer-style question directly: if a modern frontier Chinese model with carefully engineered prompting already solves the task, then claims about the necessity of local adaptation should be narrowed accordingly.

These methods are important because they show what can be achieved from symbol injection alone. In the current results, local dictionary prompting is substantially better than zero-shot generation, but it often expands compact titles into explanatory paraphrases rather than reconstructing the reference form. The frontier DeepSeek variant shows that prompt-only inference becomes much stronger once both the model and the prompt design are upgraded aggressively.

4.2 Synthetic SFT baselines

The second family uses synthetic training data. Synthetic examples are built from monolingual classical Chinese strings and a Tangut dictionary by first constructing a Chinese-character-to-Tangut-character map from dictionary explanations and then replacing characters according to one of four regimes.

- `baseline3` keeps a mixed-script input by replacing only 30–70% of replaceable Chinese characters with Tangut.
- `baseline3_1_unk` enforces pure Tangut inputs by replacing every dictionary miss with `[UNK]`.
- `baseline3_3_semantic` also enforces pure inputs, but replaces misses through vector-space semantic projection instead of `[UNK]`.
- `baseline3_2_multitask` keeps the same pure-input construction as the `UNK` branch but changes the target format to include explicit alignment scaffolding.

The multitask target is concrete rather than notional:

```
<dict_match>{Tangut input}</dict_match>
<literal>{character-level gloss proxy}</literal>
{final Chinese title}
```

This gives the model both a copy-style alignment channel and a final title-generation target in the same sequence. The central design question is therefore whether the model benefits more from input purity, approximate semantic preservation, or explicit alignment structure. At a high level, this family plays the role that synthetic parallel data often plays in low-resource MT: it converts monolingual or weakly aligned resources into supervised signals that a translation model can absorb [4]. The current evidence favors explicit alignment structure.

4.3 Preference optimization

The final family applies DPO to an SFT base using reward-defined chosen/rejected pairs [7]. Candidate preferences are generated by sampling five outputs per training prompt from the SFT model at temperature 0.8 and top-p 0.95, scoring them with

$$r(y) = \mathrm{lex}(y) - 0.01 \log \mathrm{ppl}(y),$$

where $\mathrm{lex}(y)$ is the repository's lexical-coverage score. We then keep the top-scoring output as chosen and the bottom-scoring output as rejected whenever the reward gap is at least 0.05. We study this family in two stages because the repository now contains both a failure case and a repaired variant.

The legacy pipeline first selects the best SFT base among the mixed, `UNK`, and semantic variants using chrF++, then constructs preference pairs from that base. In the current snapshot, that selector chooses the `UNK` model, yielding the legacy checkpoint `final_v2`. This run is still important because it exposes the failure mode directly: if the base model is weaker and the reward is partly misaligned, DPO can degrade title reconstruction rather than improve it.

The follow-up DPO runs start instead from the strongest supervised base, the multitask SFT model. After auditing the preference data, we remove invalid rows and keep only pairs whose reward gap exceeds a threshold, testing both a looser filter (≥ 0.2, 1,996 pairs) and a stricter filter (≥ 0.4, 498 pairs). For each filtered set we train two variants: standard sigmoid DPO and a noise-aware alternative that combines TRL's robust DPO loss with WPO-style reweighting for off-policy preferences [8, 9]. This turns the DPO family into a controlled diagnosis of three factors at once: base-model choice, pair-quality filtering, and loss robustness.

4.4 Training and judge details

All learned models start from Qwen2.5-7B-Instruct and use LoRA adapters. The SFT runs use three epochs, learning rate 2×10^{-4}, batch size 4, gradient accumulation 4, rank-64 LoRA, and maximum sequence length 512. The DPO runs use two epochs, learning rate 5×10^{-5}, batch size 1, gradient accumulation 8, $\beta = 0.1$, rank-32 LoRA, cosine decay, and maximum sequence length 512. The robust variants add label smoothing 0.1 and WPO-style weighting. The full run configurations are saved alongside each checkpoint.

The reference-aware judge is a separate supplementary evaluator, not a training signal. It uses an Azure-hosted GPT-5.4 deployment and scores each candidate on four integer dimensions: reference agreement, source faithfulness, title-style fitness, and overall quality. The judge prompt sees the Tangut source, the gold reference, and the candidate simultaneously and explicitly penalizes expansions of short titles into explanatory modern sentences. We use this judge because the repository's original reference-free judge is not strong enough to anchor the task, but we still treat it as supplemental evidence rather than a replacement for human evaluation.

5 Main Results

Table 1 now includes a much stronger prompt-only comparison. The central pattern is therefore no longer a simple "SFT good, DPO bad" split, and it is also no longer defensible to argue that stronger Chinese chain-of-thought prompting simply fails on this task. The DeepSeek-V3.2 few-shot dictionary baseline is a genuinely strong upper-bound-style prompt-only result: it reaches 28.54 chrF++, 2.62 semantic completeness, and 2.78 fluency on the legacy repository judge before later becoming the strongest non-reference system under the reference-aware judge.

The positive method result instead lies inside the local trainable regime. Structured multitask SFT remains the strongest pure supervised baseline. Once DPO is rerun on that base with reward-gap filtering, the story changes again: strict filtering does not just rescue DPO qualitatively, it produces the strongest local learned configuration in the repository. The strict gap-0.4 sigmoid DPO variant is the clearest example. Tightening the reward-gap threshold from 0.2 to 0.4 cuts the preference set from 1,996 pairs to 498, but it raises local chrF++ to 30.01, preserves 5 exact matches, and brings the average output length back near the reference ratio (1.12). This is notable because it exceeds the stronger frontier prompt-only system on chrF++ (28.54), despite using only the repository's tiny real Tangut set plus filtered synthetic preferences. The strict robust+weighted variant is slightly weaker on chrF++ (27.52) but is still much cleaner than its gap-0.2 counterpart. By contrast, the looser gap-0.2 sigmoid model already showed reference-aware adequacy gains, yet it overgenerated badly (length ratio 1.74), while the looser robust+weighted model preserved suffixes most aggressively but contaminated 7 outputs.

Method	Lex ↑	PPL ↓	chrF++ ↑	Sem ↑	Flu ↑
Prompt-only					
Zero-shot	0.1165	20.21	0.19	1.00	1.56
Dict prompt	0.8187	4474.44	6.13	2.30	2.40
Dict+CoT	0.8723	18119.16	7.59	2.30	1.88
Frontier DeepSeek	0.6234	7438.72	28.54	2.62	2.78
Local learned					
Mixed-SFT	0.3192	298684.91	17.85	1.42	1.88
UNK-SFT	0.3514	11031.18	22.11	1.66	1.96
MT-SFT	0.4839	**3322.93**	25.99	1.70	**2.26**
Sem-SFT	0.3723	17064.38	17.03	1.46	1.84
Legacy DPO	0.4224	314577.16	19.39	1.52	1.76
MT-DPO g0.2 sig	**0.5694**	303438.00	20.93	**1.88**	1.86
MT-DPO g0.2 rob	0.5249	238892.42	27.67	**1.88**	1.88
MT-DPO g0.4 sig	0.4833	9573.71	**30.01**	1.84	2.04
MT-DPO g0.4 rob	0.4547	20012.69	27.52	1.74	2.08
Human ref.	0.5733	1646139.34	100.00	2.24	2.34

Table 1: Repository-level metrics on the 50-example Tangut test set. The Frontier DeepSeek row is a proprietary prompt-only upper bound using dictionary grounding plus four development-set few-shot exemplars. Here `g0.2` and `g0.4` denote minimum preference-gap thresholds of 0.2 and 0.4. Bold marks the best local learned value in each column, excluding prompt-only baselines and the human reference. The key comparison is no longer "prompting versus training" in the abstract: frontier prompting is strongest on several reference-free metrics, but strict gap-0.4 multitask DPO still gives the best non-reference chrF++ and the strongest local raw-overlap result.

These disagreements are precisely why the paper should not rely on a single automatic metric. Raw chrF++, contamination, and the later reference-aware judge each reward different parts of the short-title problem. In the current Tangut setting, the defensible claim is comparative rather than absolutist: frontier prompting is the strongest single overall system, structured SFT is the right local starting point, and high-gap multitask-base DPO can improve on it. Loose filtering favors some adequacy gains at the cost of unstable surface behavior, while strict filtering yields the best local raw/clean trade-off.

The new statistical analysis makes this ranking more precise. Under 5,000 paired bootstrap resamples, the best local DPO model improves over multitask SFT by +4.02 chrF++ with a 95% confidence interval of [0.84, 7.80], so the local chrF++ gain is not just a single-split fluctuation. At the same time, the frontier system retains a robust advantage over that local model on reference-aware overall (-0.58 when measured as local minus frontier, 95% CI [-1.04, -0.12]) and exact match (-0.14, 95% CI [-0.26, -0.04]). By contrast, the chrF++ delta between the same two systems is not robust under bootstrap (+1.47, 95% CI [-8.09, 10.29]). In other words, the local model is stronger on one surface-overlap metric, but the frontier model remains the better single-shot choice under the more reference-aware evidence.

6 Analysis

6.1 Reference-aligned diagnostics

Repository-level metrics alone do not fully explain the new ranking. Additional diagnostics from stored predictions make the trade-off much clearer. The first important fact is that the frontier DeepSeek baseline is genuinely clean: it reaches 12/50 exact matches, no contamination, 24/27 suffix preservation, and a near-reference length ratio of 0.95. In other words, the frontier result is not just a high-capacity version of the verbose dictionary baseline. It is a strong short-title reconstruction system in its own right.

Within the local learned family, the multitask model remains the cleanest pure SFT system, with 5 exact matches, only 1 contaminated output, and a near-reference output length ratio (1.10). The looser gap-0.2 sigmoid DPO follow-up keeps the same 5 exact matches and drops contamination to 0/50, but stretches titles to 1.74 times the reference length. The looser robust+weighted follow-up instead preserves title suffixes most aggressively (23/27), but pays for that with 7 contaminated outputs.

The stricter gap-0.4 follow-ups change the surface behavior substantially. The strict sigmoid model keeps 5 exact matches, raises title-suffix preservation to 22/27, and restores the length ratio to 1.12 while reaching the highest local chrF++ in the repository (30.01). The strict robust+weighted model also keeps 5 exact matches and 22/27 suffix preservation, but with only 2 contaminated outputs instead of 7. In other words, stricter pair filtering does not merely preserve the earlier DPO gains; it regularizes them into something much closer to the short-title target distribution.

By contrast, the older local dictionary-prompting baseline often produces outputs that are too long and too explanatory. Its average output is about 1.83 times the reference length. This helps lexical coverage but hurts exact title reconstruction. The comparison between `baseline2` and the frontier DeepSeek row is therefore methodologically useful: prompt-only inference is not doomed here, but it requires a much stronger model and a much stricter title-oriented prompt than the local baseline setup.

6.2 Oracle complementarity and workflow value

The strongest new result is not another single-system win, but the size of the best-of-two upper bound. Table 3 compares the strongest frontier system against the strongest local system and then reports separate item-level oracles for exact match, chrF++, and the reference-aware overall score.

The oracle numbers are much larger than either system alone. The exact-match union rises from 12/50 for the frontier baseline and 5/50 for the best local model to 13/50 overall. More importantly, the corpus chrF++ oracle rises to 40.16 and the reference-aware overall oracle rises to 3.06. This gap is too large to dismiss as noise or a trivial restatement of the pairwise win/loss counts. It means the local model is not merely a worse copy of the frontier model; it is generating hypotheses that are often different and sometimes decisively useful. Indeed, the two systems produce identical outputs on only 4/50 items, while on the remaining 46 items the better of the two reaches mean judge overall 2.89.

This changes the practical reading of the paper. The strongest deployment claim is no longer "use the local model instead of a frontier API." It is instead "if a frontier API is available, the local model is still useful as a complementary hypothesis generator." That is a stronger scholar-facing story than simple replacement: one can generate a frontier candidate, generate a local candidate, and then either ask a judge to arbitrate or surface the disagreement to a human annotator.

Method	Exact Match	Contam. Rate	Suffix Preserve	Len. Ratio
Prompt-only				
Dict prompt	0/50	0.00	3/27	1.83
Frontier DeepSeek	12/50	0.00	24/27	0.95
Local learned				
UNK-SFT	3/50	0.06	17/27	**1.05**
MT-SFT	**5/50**	0.02	19/27	1.10
Legacy DPO	2/50	0.18	18/27	1.88
MT-DPO g0.2 sig	**5/50**	**0.00**	19/27	1.74
MT-DPO g0.2 rob	4/50	0.14	**23/27**	1.58
MT-DPO g0.4 sig	**5/50**	0.04	22/27	1.12
MT-DPO g0.4 rob	**5/50**	0.04	22/27	1.26
Human ref.	50/50	0.00	27/27	1.00

Table 2: Reference-aligned diagnostics derived from stored predictions. "Contam." denotes mixed-script or artifact contamination. In the last two columns, the table reports suffix-preservation count first and length ratio second. Bold marks the best local learned value in each column, excluding prompt-only baselines and the human reference. The frontier DeepSeek row shows that strong prompt-only inference can already reconstruct many short titles cleanly, while the new gap-0.4 local DPO variants keep most of the suffix gains of multitask-base DPO and restore near-reference length relative to their gap-0.2 counterparts.

System	Exact Match	chrF++	Overall
Frontier DeepSeek	12/50	28.54	2.80
MT-DPO g0.4 sig	5/50	30.01	2.22
Best-of-two oracle	13/50	40.16	3.06

Table 3: Complementarity between the strongest frontier and local systems. The two systems produce identical outputs on only 4/50 items, so the oracle is not just restating a near-duplicate ranking. Instead, it shows that local adaptation supplies genuinely different useful hypotheses: selecting the better output of the two systems yields large gains on both raw overlap and reference-aware utility.

6.3 A direct test of stronger CoT prompting

The obvious objection to this project is straightforward: why not simply use a stronger Chinese frontier LLM with better chain-of-thought prompting? The new DeepSeek-V3.2 few-shot baseline lets us answer that objection directly. The short answer is that strong frontier prompting is indeed very effective. On the reference-aware judge, it reaches overall 2.80, above the best local score of 2.22, and in direct pairwise comparison it beats the strict gap-0.4 sigmoid model on 20 items, ties on 23, and loses on 7.

But this does not make the local-model story empty. The frontier advantage is concentrated on canonical or bibliographically regular titles, where broad Chinese title priors help a great deal. It is exact or near-exact on cases such as "百千印陀羅尼經", "同音", "類林", and "地藏菩薩本願經", where the best local model drifts badly. At the same time, the local model still wins a non-trivial minority of hard cases that are short, elliptical, or lexically awkward: it preserves "五部經" where the frontier model normalizes it to "五類經"; it stays close to "正理滴之句義顯具" while the frontier model rewrites it as "德宜點義句顯宣論"; it translates "到賢" as "達賢" while the frontier model

Method	Ref. Agr.	Src. Faith.	Title Style	Overall
Prompt-only				
Dict prompt	1.50	1.52	2.10	1.50
Frontier DeepSeek	2.78	2.82	4.48	2.80
Local learned				
UNK-SFT	1.78	1.84	4.36	1.82
MT-SFT	1.92	1.98	4.18	1.98
Legacy DPO	1.64	1.72	3.80	1.68
MT-DPO g0.2 sig	2.08	2.24	4.14	**2.22**
MT-DPO g0.2 rob	2.10	2.16	3.78	2.12
MT-DPO g0.4 sig	**2.12**	**2.26**	4.60	**2.22**
MT-DPO g0.4 rob	2.10	2.20	**4.70**	**2.22**
Human ref.	5.00	5.00	5.00	5.00

Table 4: Reference-aware judge scores on a 1–5 scale. The judge sees the Tangut input, the gold reference, and the candidate output simultaneously. Bold marks the best local learned value in each column, excluding prompt-only baselines and the human reference. The frontier DeepSeek baseline is the strongest non-reference system overall, but within the local family the best overall score is shared by three multitask-base DPO variants; the stricter gap-0.4 runs win the local tie-break dimensions, with MT-DPO g0.4 sig leading on reference agreement and source faithfulness and MT-DPO g0.4 rob leading on title-style fitness.

hallucinates “五台山觀音普賢殿寺”; and it exactly recovers “番言金剛王乘根” while the frontier model collapses to “番”.

The comparison therefore sharpens the paper rather than invalidating it. A paper claiming that “stronger CoT still cannot do the task” would now be wrong. A stronger and more honest claim is that frontier prompting is a proprietary upper bound, while structured local adaptation remains scientifically useful because it changes the error regime, remains trainable and reproducible, and supplies complementary hypotheses that substantially enlarge the best-of-two upper bound.

6.4 Reference-aware judging changes the ranking

The legacy LLM judge in the repository is reference-free, so we reran evaluation with a stricter judge that sees the Tangut input, the gold reference, and the candidate output simultaneously. This rerun serves two purposes. First, it checks whether the human reference is correctly ranked when the judge has the information it should need. Second, it tests whether the raw-metric ranking survives once the judge can penalize explanatory overgeneration and content drift directly.

Table 4 changes the paper narrative once more. The human reference still receives perfect 5.0 scores on all dimensions, so the judge is well anchored. Among non-reference systems, the frontier DeepSeek baseline is now the clear winner at 2.80 overall. Within the local learned family, however, the best overall score is shared by three multitask-base DPO variants: the loose sigmoid run, the strict sigmoid run, and the strict robust+weighted run all reach 2.22. Within that local tie, the strict gap-0.4 runs are stronger on title reconstruction: the strict sigmoid model reaches the highest local reference-agreement score (2.12) and the highest local source-faithfulness score (2.26), while the strict robust+weighted model reaches the highest local title-style score (4.70). The older `final_v2` checkpoint remains worse than `baseline3_2_multitask`, so the positive DPO result is specific to the stronger base and higher-quality pairs, not to DPO in the abstract.

The subset breakdown makes the new trade-off sharper. On the 27 reference titles containing

Comparison ($B - A$)	Δ chrF++	Δ Overall	Δ Exact
Local – Frontier	+1.47 [−8.09, 10.29]	−0.58 [−1.04, −0.12]	−0.14 [−0.26, −0.04]
Local – MT-SFT	+4.02 [0.84, 7.80]	+0.24 [0.00, 0.46]	0.00 [0.00, 0.00]
Legacy DPO – MT-SFT	−6.60 [−11.84, −1.73]	−0.30 [−0.64, 0.04]	−0.06 [−0.14, 0.00]

Table 5: Paired bootstrap comparisons with 5,000 resamples. "Overall" denotes the mean reference-aware judge score, and "Exact" denotes exact-match rate. The table shows that the local chrF++ gain over multitask SFT is robust, while the frontier system retains a robust advantage over the best local model on reference-aware overall and exact match.

common title suffixes, the frontier model reaches mean overall score 3.15, well above the best local runs (2.41). On the 23 non-suffix items, however, the gap narrows: the frontier model reaches 2.39, while the looser `final_gap02_multitask_sigmoid` reaches 2.30. Within the local family, the strict gap-0.4 variants dominate the suffix-bearing subset, clearly above `baseline3_2_multitask` (2.15), and they do so without the earlier surface penalties: the strict sigmoid run stays near reference length (1.12) and the strict robust run drives contamination back to 0.0% on this subset. On the non-suffix items, the looser gap-0.2 sigmoid variant remains strongest among local models. This is strong evidence that prompt-only reasoning and local parameter adaptation are exploiting different kinds of regularity.

6.5 Human-reference anchor

One of the strongest pieces of evidence in the repository is the `human_reference` experiment. The gold reference receives perfect chrF++ and perfect reference-aware judge scores, as expected, but only moderate lexical coverage and catastrophic perplexity. This reveals a core evaluation issue: the current lexical coverage metric measures overlap with dictionary-derived semantics rather than agreement with the gold title, while the perplexity model penalizes compact title style and historical transliterations.

The implication is methodological, not cosmetic. If we were to select models purely by lexical coverage or perplexity, we would risk preferring systems that paraphrase, overgenerate, or stop poorly instead of systems that best reconstruct the true title. The new frontier baseline makes this even clearer: it is strong overall under the reference-aware judge, but not because it wins lexical coverage or perplexity outright. Conversely, the strict gap-0.4 sigmoid model reaches the best non-reference chrF++ while remaining far below the frontier model on the reference-aware judge. Therefore, in the paper, lexical coverage and perplexity should be reported as supporting diagnostics, not as the primary decision criteria.

6.6 Statistical robustness and data hygiene

The 50-example test set is too small to justify raw point estimates without uncertainty. The bootstrap results in Table 5 clarify which claims are stable. The local chrF++ gain from multitask SFT to strict gap-0.4 sigmoid DPO survives resampling, with a 95% confidence interval entirely above zero. The frontier model's advantage over the best local model on reference-aware overall and exact match also survives resampling. By contrast, the chrF++ difference between those two systems does not: its 95% interval crosses zero widely. This is exactly the kind of mixed picture that the paper should surface rather than hide. The correct interpretation is therefore not "local beats frontier" or "frontier dominates local" in the abstract, but "different metrics support different deployment decisions, while the oracle shows the systems are complementary."

Audit item	Value
Total preference pairs	3867
Duplicate chosen/rejected pairs	3
Non-finite reward entries	3
Mean reward gap	0.2396
Chosen closer to gold proxy	71.3%
Chosen closer on gap $(0.05, 0.10]$	62.3%
Chosen closer on gap $(0.20, 0.40]$	75.4%
Chosen closer on gap ≥ 0.40	81.5%
Pairs kept with gap ≥ 0.20	1996
Pairs kept with gap ≥ 0.30	1010
Pairs kept with gap ≥ 0.40	498

Table 6: Lightweight audit of the current DPO pair construction pipeline. Higher reward gaps correlate with cleaner preference labels under a simple gold-proxy check, which helps explain why strict filtering stabilizes the later multitask-base DPO runs.

The overlap audit addresses a separate reviewer concern. There is zero exact source, target, or source-target-pair overlap across the train/dev/test splits; zero exact overlap between test references and the 50k synthetic multitask targets; and zero exact overlap between the four frontier few-shot exemplars and the test references. Real-split character n-gram overlap is non-zero, which is expected in a domain where many titles share Buddhist morphemes and suffixes, but the synthetic multitask set is not memorizing the held-out titles: its mean maximum Jaccard overlap with test references is only 0.027 for character bigrams, 0.004 for trigrams, and 0.000 for 4-grams. That does not prove complete immunity from leakage in a broader sense, but it does rule out the simplest exact-copy explanation for the current results.

6.7 Legacy DPO fails for different reasons than multitask-base DPO

The old negative result is not a random bad run. The preference data itself is imperfect. The repository contains 3,867 preference pairs, including a small number of duplicate or invalid entries. More importantly, a quick audit against the original synthetic targets shows that the reward-chosen candidate is closer to gold only about 71% of the time under a simple similarity proxy, and that this quality rises monotonically with reward gap: 62.3% for gaps in $[0.05, 0.10]$, 66.3% for $(0.10, 0.20]$, 75.4% for $(0.20, 0.40]$, and 81.5% for gaps ≥ 0.40. In other words, the high-gap filter is not a cosmetic hyperparameter. It is directly isolating cleaner preference labels.

The legacy selector also excludes the strongest multitask model from the DPO-base search. That turns out to matter. Once we switch to the multitask base and keep only pairs with reward gap ≥ 0.2, DPO stops being a blanket negative result and begins to beat the multitask SFT baseline under the reference-aware judge. Tightening the filter further to gap ≥ 0.4 makes the mechanism much clearer. Relative to the loose gap-0.2 sigmoid model, the strict sigmoid model raises chrF++ from 20.93 to 30.01 and shrinks the length ratio from 1.74 to 1.12 while preserving the same 2.22 overall judge score. Relative to the loose robust+weighted model, the strict robust+weighted model reduces contamination from 7/50 to 2/50 and keeps the overall judge score at 2.22. Even a deterministic short-title normalization pass raises `final_gap02_multitask_robustwpo` only to chrF++ 29.06, still below the raw strict-sigmoid result. The reward is therefore not useless, but low-gap pairs inject exactly the kind of surface noise that a short-title system can least afford.

7 Discussion and Limitations

The obvious deployment question is now easier to answer than before. If a user can rely on a proprietary frontier API and wants the strongest current single inference-time result on this test set, the DeepSeek-V3.2 few-shot baseline is the best system we measured. The paper should say that directly. The contribution is not that prompt-only frontier models fail.

The contribution is instead threefold. First, the paper shows how far an open, local, trainable pipeline can be pushed with only 491 real pairs: structured multitask SFT plus high-gap DPO yields a model that surpasses the frontier prompt-only baseline on chrF++ and wins 7/50 pairwise judge cases. Second, the frontier comparison exposes complementary failure modes rather than simple domination, and the best-of-two oracle is large enough to make hybrid or scholar-in-the-loop use realistic. Third, the paper shows that reference-free metrics and noisy preference construction can badly distort conclusions in this task. Those three points remain valuable even when a proprietary prompt-only upper bound is stronger overall.

The paper should also be explicit about scope. The current dataset supports Tangut short-text translation, especially title-like strings, not general open-domain Tangut sentence translation. The subset analysis strengthens that boundary in a slightly different way than before: the largest local gains from strict gap-filtered DPO are concentrated on reference titles with canonical suffix patterns, while the looser gap-0.2 sigmoid variant remains stronger on the non-suffix subset. That scope is still meaningful: many historically important Tangut materials appear precisely in compact catalog, title, or scholastic forms, and low-resource methods for such data are valuable.

Two limitations matter most for future work. First, the real dataset is extremely small, so the paper cannot claim stable behavior beyond the current narrow domain. Second, the current paper still stops one step short of the strongest workflow claim: we show a large frontier-plus-local oracle, but we do not yet train or prompt a real adjudicator that closes that gap automatically, nor do we provide a human validation study for the reference-aware judge. The most useful next experiments are therefore not "more DPO" in the abstract, but hybrid adjudication, frontier-to-local distillation, pair-selection curricula, title-aware decoding control, and training objectives that reward both adequacy and clean title form directly.

8 Conclusion

The current repository supports a stronger paper than before, but the final message is no longer "SFT good, DPO bad," and it is also not "frontier prompting fails." A carefully prompted DeepSeek-V3.2 baseline is the strongest non-reference system overall, so the paper should treat it as a proprietary upper bound rather than deny its strength. The more useful scientific result is narrower and more actionable: within the local trainable regime, structured multitask supervision remains the strongest pure SFT baseline, and stricter gap-filtered DPO produces the strongest local model. `final_gap04_multitask_sigmoid` reaches the highest local chrF++ (30.01) and ties for the best local reference-aware overall score (2.22), while `final_gap04_multitask_robustwpo` reaches the same overall score with the strongest local title-style fitness. The legacy `final_v2` checkpoint remains a genuine negative control, so the positive DPO result is conditional on both the base model and the pair filter. More importantly, the frontier-local oracle rises to 40.16 chrF++, 3.06 reference-aware overall, and 13 exact matches, which shows that local adaptation contributes complementary hypotheses even when a proprietary frontier system is stronger overall. Several reference-free metrics also remain misaligned enough to mis-rank the human reference. Together, these findings argue for a more precise lesson: in ultra-low-resource Tangut short-text translation, frontier prompting is a strong upper bound, explicit structural supervision is the right local base recipe, aggressively filtered preference optimization can improve on it, and the most promising next step is not replacement but hybrid use under reference-aware evaluation.

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